

Given a CCS process P and a finite set of channel names $L \subseteq \mathcal{A}$, define the operator τ_L that *hides* every action on a channel in L by replacing it with a silent τ step. Formally, τ_L is characterized by:

$$\frac{P \xrightarrow{\alpha} P'}{\tau_L(P) \xrightarrow{\tau} \tau_L(P')} \quad \alpha \in L \cup \bar{L} \quad \frac{P \xrightarrow{\alpha} P'}{\tau_L(P) \xrightarrow{\alpha} \tau_L(P')} \quad \alpha \notin L \cup \bar{L}$$

Show that there exists a CCS context $C_L[\cdot]$ such that:

$$\tau_L(P) \sim C_L[P] \quad \text{for all CCS processes } P$$

where $\sim$ denotes weak bisimilarity.

The context requires every visible action $a \in L$ to be a τ , using only CCS. The only source of τ in CCS is the COM rule: two parallel processes synchronize on complementary channels a and $\bar{a}$, producing τ .

1. Rename every $a \in L$ to a fresh private channel b_a . This gets L out of the way without blocking it.

$$C_L[P] = P^*$$

where

$$P^* = P[b_a/a \mid a \in L]$$

2. Run in parallel a partner that is *always* ready to synchronize on b_a , in either direction, and for each $a \in L$.

$$C_L[P] = P^* \mid A_L$$

where

$$A_L = \prod_{a \in L} A_a$$

and

$$A_a \stackrel{\text{def}}{=} b_a.A_a + \bar{b}_a.A_a$$

3. Restrict on $\{b_a\}$ to make b_a invisible from the outside.

$$C_L[P] = (P^* \mid A_L) \setminus \{b_a \mid a \in L\}$$

Claim. For all $P \in \text{CCS}$, $\tau_L(P) \sim C_L[P]$.

Proof. Two processes are weakly bisimilar if every transition of one can be matched by the other, up to silent τ steps, and the resulting states remain related indefinitely. Rather than reasoning about $\sim$ abstractly, we certify this directly: it suffices to exhibit a concrete relation R containing the pairs we wish to relate and verify that R itself has the mutual-matching property $(*)$ below. Since $\sim$ is the largest such relation, any verified R is contained in $\sim$, which gives us our conclusion.

We therefore take

$$R = \{(\tau_L(P), C_L[P]) \mid P \in \text{CCS}\} \cup \sim$$

The $\cup \{\sim\}$ is a technical convenience: after each matching step the residual pair need only land back in R , and including $\sim$ means any pair already known to be bisimilar closes the check immediately without infinite regression.

Formally, $\sim$ is characterized by: for all processes P, Q ,

$$P \sim Q \iff \begin{array}{l} \text{if } P \xrightarrow{\alpha} P' \text{ then } \exists Q'. Q \xRightarrow{\alpha} Q' \text{ and } P' \sim Q' \\ \text{if } Q \xrightarrow{\alpha} Q' \text{ then } \exists P'. P \xRightarrow{\alpha} P' \text{ and } P' \sim Q' \end{array} \quad (*)$$

where $\xRightarrow{\alpha}$ denotes weak transition (zero or more τ steps, then α , then zero or more τ steps; for $\alpha = \tau$ this includes doing nothing).

We now show R satisfies $(*)$.

Take any pair $(\tau_L(P), C_L[P]) \in R$.

- **Direction (1): suppose** $\tau_L(P) \xrightarrow{\mu} \tau_L(P')$.

By the defining rules of τ_L there are two cases:

- *Case* $\mu = \tau$, $\alpha \in L$: Renaming gives $P^* \xrightarrow{b_a} P'^*$; the right branch of A_a provides $\bar{b}_a$:

$$\frac{\frac{P \xrightarrow{a} P'}{P^* \xrightarrow{b_a} P'^*} \text{Ren} \quad \frac{\frac{\frac{}{\bar{b}_a \cdot A_a \xrightarrow{\bar{b}_a} A_a} \text{Act}}{\bar{b}_a \cdot A_a \xrightarrow{\bar{b}_a} A_a} \text{Sum}}{A_a \xrightarrow{\bar{b}_a} A_a} \text{Par-L}}{A_L \xrightarrow{\bar{b}_a} A_a \mid A_{L \setminus \{a\}}} \text{Com} \quad \frac{P^* \mid A_L \xrightarrow{\tau} P'^* \mid A_L}{C_L[P] \xrightarrow{\tau} C_L[P']} \text{Res}$$

So $C_L[P] \xrightarrow{\tau} C_L[P']$ and $(\tau_L(P'), C_L[P']) \in R$. The case $\alpha = \bar{a}$ is symmetric: renaming gives

$P^* \xrightarrow{\bar{b}_a} P'^*$ and the left branch of A_a provides the complementary b_a ; the same tree applies.

- *Case* $\mu = \alpha \notin L \cup \bar{L}$: Renaming leaves α unchanged and restriction does not block it:

$$\frac{\frac{\frac{P \xrightarrow{\alpha} P'}{P^* \xrightarrow{\alpha} P'^*} \text{Ren}}{P^* \mid A_L \xrightarrow{\alpha} P'^* \mid A_L} \text{Par-L}}{C_L[P] \xrightarrow{\alpha} C_L[P']} \text{Res, } \alpha \notin \{b_a \mid a \in L\}$$

So $C_L[P] \xrightarrow{\alpha} C_L[P']$ and $(\tau_L(P'), C_L[P']) \in R$.

- **Direction (2): suppose** $C_L[P] \xrightarrow{\mu} C_L[P']$.

Every visible transition must originate from P^* . A_L alone produces no τ : it has no τ -prefixed subterm and cannot self-synchronize, so any τ arises only by synchronizing with P^* on a complementary b_a channel. Restriction removes all observable b_a -labelled moves from the composed system, but does not prevent the internal synchronization that produces τ .

- *Case* $\mu = \tau$: The τ arises in exactly two ways.

Sub-case: sync on b_a ($P \xrightarrow{a} P'$, $a \in L$):

$$\frac{P \xrightarrow{a} P'}{\tau_L(P) \xrightarrow{\tau} \tau_L(P')} \quad a \in L, \text{ first rule of } \tau_L$$

The case $P \xrightarrow{\bar{a}} P'$ is symmetric.

Sub-case: silent step ($P \xrightarrow{\tau} P'$):

$$\frac{P \xrightarrow{\tau} P'}{\tau_L(P) \xrightarrow{\tau} \tau_L(P')} \quad \tau \notin L \cup \bar{L}, \text{ second rule of } \tau_L$$

In both sub-cases $\tau_L(P) \xrightarrow{\tau} \tau_L(P')$ and $(\tau_L(P'), C_L[P']) \in R$.

- *Case* $\mu = \alpha \notin L \cup \bar{L}$: Then $P \xrightarrow{\alpha} P'$ (renaming leaves α unchanged and restriction does not block it):

$$\frac{P \xrightarrow{\alpha} P'}{\tau_L(P) \xrightarrow{\alpha} \tau_L(P')} \quad \alpha \notin L \cup \bar{L}, \text{ second rule of } \tau_L$$

So $\tau_L(P) \xrightarrow{\alpha} \tau_L(P')$ and $(\tau_L(P'), C_L[P']) \in R$.

In all cases every condition of $(*)$ is satisfied, so R is a weak bisimulation. Since $\sim$ is the largest weak bisimulation, every bisimulation is contained in $\sim$, so $R \subseteq \sim$. As $(\tau_L(P), C_L[P]) \in R$, we conclude $\tau_L(P) \sim C_L[P]$ for all P . $\square$